energy. When the accelerated expansion of the universe was discovered in 1998, the new paradigm placed us squarely in the path of a dark-energy-dominated future: one in which the cosmos gets progressively emptier, colder, and darker until all structure decays and we reach the ultimate Heat Death. But this is just an extrapolation, one that's predicated on dark energy being an unchanging cosmological constant. As we've seen, if whatever is responsible for cosmic acceleration falls into the category of phantom dark energy, or if it somehow changes over time, the implications for the cosmos are drastically different.

Unfortunately, as far as observations are concerned, dark energy doesn't give us a lot to hold on to. It is, as far as we can tell, invisible, undetectable in laboratory experiments, completely uniformly distributed through space, and only really noticeable at all by its indirect effects over scales much larger than our galaxy.

Generally speaking, there are two things we can measure. The first is the expansion history of the universe, which at the moment we study primarily by looking at very distant supernovae and figuring out how fast they're receding. The other is the history of the formation of structure, where by "structure" we are generally referring to galaxies and clusters of galaxies, because all the little things like stars and planets are just annoying details if you're a cosmologist. Measuring this is a bit less straightforward, but also allows for a lot of creative uses of massive piles of data. The trick is to get images and spectra of as many galaxies as possible, over a giant volume of space (and a large patch of cosmic history), and use statistical methods to infer how all that matter came together over time. Together, these two kinds of measurements can tell us how the space-stretching properties of dark energy have affected the universe as a whole and how much it's impeded the efforts of matter to clump together and form things like galaxies and clusters and us.

When you only have two things you can measure to determine the WHOLE FATE OF THE UNIVERSE, it makes sense to invest a lot in measuring them very, very well. There's been a surge of interest in the last couple decades in new telescopes and surveys with "dark energy" prominently featured in their science cases. Some are designed around the promise of how well they can use expansion and structure growth measurements to determine the dark energy equation of state parameter w (discussed in Chapter 5). If w = -1 exactly, now and in the past, we have a